Supply Chain Analysis Using SQL

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Project Overview

This project focuses on analyzing supply chain data using SQL to evaluate supplier performance, cost efficiency, defect rates, and operational bottlenecks. The goal is to derive actionable insights that can support better business decisions in supply chain management.

Objectives

- Identify top-performing and high-risk suppliers
- Analyze revenue contribution across products
- Evaluate cost efficiency and operational delays
- Detect quality issues using defect and inspection data

Dataset Description

Table Name: `supply_chain`

Key Columns:

- `product_type` – Category of product
- `supplier_name` – Supplier providing goods
- `price` – Revenue generated
- `manufacturing_cost` – Production cost
- `shipping_cost` – Logistics cost
- `defect_rate` – Quality indicator
- `inspection_result` – Pass/Fail status
- `lead_time` – Delivery time
- `production_volumes` – Quantity produced

SQL Analysis

Total Revenue by Product

Business Question:
Which product category generates the highest revenue?

```
SELECT product_type,
       SUM(price) AS total_revenue
FROM supply_chain
GROUP BY product_type
ORDER BY total_revenue DESC;
```

Insight:
Skincare products generate the highest revenue, making them the primary business driver.

Supplier Contribution

Business Question:
Which suppliers contribute the most to production?

```
SELECT supplier_name,
       SUM(production_volumes) AS total_production
FROM supply_chain
GROUP BY supplier_name
ORDER BY total_production DESC;
```

Insight:
Supplier 1 contributes the highest production volume, indicating dependency on a single supplier.

Defect Analysis

Business Question:
Which supplier has the highest defect rate?

```
SELECT supplier_name,
       AVG(defect_rate) AS avg_defect_rate
FROM supply_chain
WHERE inspection_result = 'Fail'
GROUP BY supplier_name
ORDER BY avg_defect_rate DESC;
```

Insight:
Supplier 1 shows the highest defect rate among failed inspections, highlighting quality concerns.

Cost vs Revenue Efficiency

Business Question:
Which supplier is least efficient?

```
SELECT supplier_name,
       SUM(price) AS total_revenue,
       SUM(manufacturing_cost + shipping_cost) AS total_cost,
       SUM(price) - SUM(manufacturing_cost + shipping_cost) AS profit
FROM supply_chain
GROUP BY supplier_name
ORDER BY profit ASC;
```

Insight:
Supplier 4 has lower profitability due to higher costs, indicating inefficiency.

High-Risk Suppliers

Business Question:
Identify suppliers with high defect rates and high lead times.

```
SELECT supplier_name,
       AVG(defect_rate) AS avg_defect,
       AVG(lead_time) AS avg_lead_time
FROM supply_chain
GROUP BY supplier_name
HAVING AVG(defect_rate) > (SELECT AVG(defect_rate) FROM supply_chain)
   AND AVG(lead_time) > (SELECT AVG(lead_time) FROM supply_chain);
```

Insight:
Suppliers identified here pose operational and quality risks simultaneously.

Top Supplier per Product

Business Question:
Which supplier performs best for each product category?

```
SELECT product_type, supplier_name, total_revenue
FROM (
    SELECT product_type,
           supplier_name,
           SUM(price) AS total_revenue,
           RANK() OVER (PARTITION BY product_type ORDER BY SUM(price) DESC)
AS rnk
    FROM supply_chain
    GROUP BY product_type, supplier_name
) t
```

```
WHERE rnk = 1;
```

Insight:
Each product category has a different top-performing supplier, indicating specialization.

Supplier Contribution Percentage

Business Question:
What percentage of total production does each supplier contribute?

```
SELECT supplier_name,
       SUM(production_volumes) * 100.0 /
       (SELECT SUM(production_volumes) FROM supply_chain) AS
contribution_pct
FROM supply_chain
GROUP BY supplier_name
ORDER BY contribution_pct DESC;
```

Insight:
Top suppliers contribute a large share, indicating supply concentration risk.

Key Business Insights

- Supplier 1 has the highest contribution but also high defect rates, indicating a quality-risk trade-off.
- Skincare products dominate revenue generation.
- Supplier 4 shows cost inefficiency with lower profitability.
- Certain suppliers exhibit both high defect rates and longer lead times, making them high-risk.
- Supply chain is dependent on a few key suppliers, increasing operational risk.

Conclusion

This SQL-based analysis highlights key inefficiencies and risks in the supply chain. By identifying high-cost suppliers, defect-prone operations, and dependency patterns, businesses can take strategic actions to optimize performance and reduce risks.

Future Improvements

- Integrate real-time data pipelines
- Build dashboards using Power BI or Tableau
- Apply predictive analytics for demand forecasting

- Optimize supplier selection using advanced models